

Group C Remaining — Network Simulation + Screenshot-on-Failure Design

Status: Design approved (operator, 2026-05-05 — "all good" on the A.3 / B.1 / C.2 recommendations after Phase 5 SSH/QMP impls landed).
Implementation pending — subagent-driven execution. Branch: Containers lava-pin/2026-05-07-pkg-vm (extends the existing in-flight branch).

Forensic anchor. The Group B-pkg-vm closure JSON listed three Group C candidates: AVD image-cache management (delivered via the bundled pkg/vm + image-cache spec, closing §6.K-debt), network simulation per AVD/VM, and hardware-screenshot capture per row. The first landed in a separate cycle; the other two are bundled here as the final piece of Group C.

1. Scope

1.1 In scope

- **Network conditions per row** (component A.3 — profile + custom override). 8 named profiles map to (downKbps, upKbps, latencyMS, lossPercent) tuples; CLI override flags supersede the profile when set. Applied AFTER guest boot completes, BEFORE the test/script runs.
- **Screenshot capture on test failure** (component B.1). When `row.Passed == false`, capture a screenshot of the guest's framebuffer and save it under `<evidence-dir>/<row-id>/screenshot-on-failure.{png,ppm}`.
- **Both pkg/emulator AND pkg/vm coverage** (component C.2 — symmetric). Same field names + flags + behavior across both packages.
- **CLI surfaces.** Both `cmd/emulator-matrix` and `cmd/vm-matrix` gain the new flags. Lava-side wrappers (`scripts/run-emulator-tests.sh`, `scripts/run-vm-{signing,distro}-matrix.sh`) forward them.

1.2 Out of scope (deferred)

- Real-time network condition changes mid-test (e.g., simulate "wifi drop at second 30"). Static-for-the-test conditions only.
- Always-capture screenshots (B.2/B.3) — only on-failure for v0.1.
- PCAP packet capture per row.
- Per-row video recording.
- DNS / DHCP / VPN simulation beyond bandwidth/latency/loss.

1.3 Non-goals

- **Not** a re-design of any existing matrix-runner internals — both new features integrate via small additions to runOne (between WaitForBoot/WaitForReady and RunInstrumentation/Run for network apply; after Run for screenshot).
- **Not** a relaxation of any existing clause. Anti-bluff posture applied: every test asserts on user-visible / forensic state (the actual bandwidth/latency/loss applied; the actual screenshot file existing on disk with non-empty bytes).

2. Architecture

Containers (branch lava-pin/2026-05-07-pkg-vm – extends current HEAD)

```
pkg/emulator/
  types.go          + NetworkConditions{ DownKbps, UpKbps,
LatencyMS, LossPercent }
                    + MatrixConfig.NetworkProfile string
                    + MatrixConfig.NetworkOverride
NetworkConditions
                    + MatrixConfig.CaptureScreenshotOnFailure
bool
                    + AVDRow.NetworkProfile string (recorded in
attestation)
                    + AVDRow.ScreenshotPath string (relative to
evidence dir; empty if not captured)
  network.go        (new) – LookupNetworkProfile,
mergeNetworkConditions(profile, override),
                    applyNetworkConditions(ctx, executor, serial,
conditions) via adb emu console
                    `network speed <kbps>` and `network delay
<ms>` commands.
  screenshot.go     (new) – CaptureScreenshot(ctx, executor,
serial, dstPath) via
                    `adb -s <serial> exec-out screencap -p`
writing PNG bytes.
  matrix.go         runOne integration: applyNetworkConditions
after WaitForBoot, CaptureScreenshot when row.Passed == false
(gated on          config.CaptureScreenshotOnFailure).
  network_test.go   (new) + screenshot_test.go (new) – fakes-
driven unit tests.

pkg/vm/
  types.go          analogous fields on VMMatrixConfig +
VMTestResult
  network.go        (new) – applyNetworkConditionsVM via SSH-in
`tc qdisc add dev eth0
                    root netem delay <ms>ms rate <kbps>kbit loss
<%>%` after
                    WaitForReady. Network interface name resolved
```

via SSH `ip route`
 screenshot.go | awk '/^default/{print \$5}'`.
 (new) – CaptureScreenshotVM via QMP
 `screendump <path>` to a
 guest-side temp path; QMP returns; then SCP-
 download from guest.
 QEMU's screendump produces PPM (no native
 PNG; operator can
 convert via ImageMagick / eog handles PPM).
 matrix.go runOne integration: same shape as pkg/
 emulator.
 network_test.go (new) + screenshot_test.go (new) – fakes-
 driven unit tests.

cmd/emulator-matrix/main.go + cmd/vm-matrix/main.go

Both gain:

--network-profile <name> one of: edge, 2g, 3g,
 4g, lte, wifi, ethernet, none
 --network-bandwidth-down <kbps> override
 --network-bandwidth-up <kbps> override
 --network-latency <ms> override
 --network-loss <%> override (float,
 0.0-100.0)
 --capture-screenshot-on-failure bool, default true

Lava parent (branch master)

scripts/run-emulator-tests.sh
 scripts/run-vm-signing-matrix.sh
 scripts/run-vm-distro-matrix.sh
 forward the new flags 1:1 (existing extra_args pattern).

No new tests required at Lava level – the contract is "flags
 forwarded".

The Containers tests cover behavior.

Network profile values

```
var networkProfiles = map[string]NetworkConditions{
  "edge": {DownKbps: 240, UpKbps: 200, LatencyMS:
    840, LossPercent: 1.0},
  "2g": {DownKbps: 50, UpKbps: 50, LatencyMS:
    500, LossPercent: 2.0},
  "3g": {DownKbps: 1500, UpKbps: 750, LatencyMS:
    100, LossPercent: 0.5},
  "4g": {DownKbps: 6000, UpKbps: 1500, LatencyMS:
    50, LossPercent: 0.1},
  "lte": {DownKbps: 12000, UpKbps: 3000, LatencyMS:
    20, LossPercent: 0.05},
  "wifi": {DownKbps: 50000, UpKbps: 10000, LatencyMS:
    5, LossPercent: 0.01},
  "ethernet": {DownKbps: 100000, UpKbps: 100000, LatencyMS:
    1, LossPercent: 0.0},
```

```

    "none":    {}, // no shaping; LookupNetworkProfile returns
                zero-value
}

```

Override merging: any non-zero override field wins over the profile's value. none profile + zero overrides = no shaping applied (early return).

3. Data flow

Successful gating run

```

operator: ./scripts/run-emulator-tests.sh \
          --network-profile 4g \
          --network-loss 0.5 \
          ...

```

```

scripts/run-emulator-tests.sh
└─ forwards --network-profile 4g --network-loss 0.5 to cmd/
emulator-matrix

```

```

cmd/emulator-matrix
└─ for each AVD:
    │   └─ Boot → WaitForBoot (Android up)
    │   └─ applyNetworkConditions(serial, conditions) – adb emu
network speed/delay
    │   └─ Install APK
    │   └─ RunInstrumentation
    │   └─ if !row.Passed && CaptureScreenshotOnFailure:
    │       └─ CaptureScreenshot(serial, evidence-dir/<avd>/
screenshot-on-failure.png)
    │   └─ append AVDRow with NetworkProfile + (if captured)
ScreenshotPath
    └─ Teardown
        └─ writeAttestation includes per-row network_profile +
screenshot_path fields

```

scripts/tag.sh (unchanged) – gates on existing schema; the 2 new fields are additive and don't affect the Group B gates.

4. Error handling

Scenario	Behavior
Unknown --network-profile <name>	exit 2 with "unknown network profile: %s" + list of valid names
applyNetworkConditions fails (adb/SSH error)	row's BootError records "network conditions apply failed: "; row Passed=false
CaptureScreenshot fails (adb error / screendump error)	log warning to stderr; row.ScreenshotPath stays empty; row.Passed signal NOT

Scenario	Behavior
	changed (screenshot is forensic-extra, not gating)
Screenshot disk-write fails	same — log + skip; non-fatal
--network-bandwidth-down <kbps> with negative value	exit 2 at flag-parse time
--network-loss <%> outside [0, 100]	exit 2
--capture-screenshot-on-failure=false with row.Passed=false	no screenshot written (operator opted out)
Profile + override merge produces zero-everything	no shaping applied (early return; this is the none profile path)

5. Testing

Test	Asserts
TestLookupNetworkProfile_Known	each of 8 profile names is expected (DownKbps, LatencyMS, LossPercentage)
TestLookupNetworkProfile_Unknown_ReturnsError	"unknown network profile"
TestMergeNetworkConditions_OverrideWinsPerField	profile=4g (down=6000, up=1000, latency=50, loss=0.1%) override.DownKbps=12000, UpKbps=2000, LatencyMS=100, LossPercentage=0.5% result.DownKbps=12000, UpKbps=2000, LatencyMS=100, LossPercentage=0.5% fields inherit from profile
TestApplyNetworkConditions_NoOpOnZeroes	conditions all-zero → no network commands N
TestApplyNetworkConditions_AdbEmuConsoleCommandsIssued_Android	conditions=4g profile fakeExecutor records console "network 6000:1500" + network 50 (or analogous emulator invocations)
TestApplyNetworkConditionsVM_TCQdiscIssued	conditions=4g → fake session records tc qdisc dev eth0 root netem 50ms rate 6000kbit 0.1%
TestCaptureScreenshot_Android_WritesPNGBytes	adb screencap fake produces bytes; CaptureScreenshot writes them to dst path
TestCaptureScreenshot_AndroidAdbError_LogsWarningReturnsNil	when adb fails, function (forensic-only) and does not propagate the error to caller
TestCaptureScreenshotVM_QMPScreendumpIssued	QMP fake records screendump <path> command

Test	Asserts
TestRunOne_NetworkConditionsAppliedBetweenBootAndTest	runOne with --network → apply happens after WaitForBoot, BEFORE RunInstrumentation
TestRunOne_ScreenshotCapturedOnRowFailure	runOne with row.Pass CaptureScreenshotOn → screenshot file exists expected path
TestRunOne_ScreenshotSkippedOnRowSuccess	runOne with row.Pass screenshot NOT captured
TestRunOne_ScreenshotSkippedWhenFlagFalse	runOne with row.Pass CaptureScreenshotOn → screenshot NOT captured

≥4 falsifiability rehearsals across the test suite.

6. Order of operations

Phase	Branch	Files
A. Containers code	lava-pin/ 2026-05-07- pkg-vm (extends)	pkg/emulator/ {types,network,screenshot,matrix,network_test,screenshot_ pkg/vm/ {types,network,screenshot,matrix,network_test,screenshot_ cmd/{emulator,vm}-matrix/main.go
B. Lava parent	master	scripts/run-emulator-tests.sh, scripts/run-vm-{signing,disc submodules/containers gitlink bump

Total: **1 Containers commit + 1 Lava parent commit**. Subagent-driven; 4-mirror push verified live.

7. Constitutional bindings

- **Sixth Law clauses 1-5.** Tests assert on user-visible behavior: the actual adb/tc commands issued, the actual screenshot file's existence + non-empty bytes, the actual network_profile field in the attestation row.
- **Clause 6.I.** Per-row schema gains network_profile + screenshot_path; tag.sh's existing Group B gates are unchanged (new fields are additive).
- **Clause 6.J/6.L.** Anti-bluff posture: screenshot-on-failure is forensic gold for diagnosing broken tests; the network-conditions apply is observable on the emulator console / via tc inside the guest. No silent failures.
- **Clause 6.M.** No new host-stability concerns. tc qdisc runs INSIDE the guest, not on the host.
- **Clause 6.N.** The Containers commit's body carries ≥4 falsifiability rehearsals per the stricter §6.N variant.
- **Decoupled Reusable Architecture.** Network profiles + screenshot capture are generic emulator/VM concerns; they live in Containers. Lava-side scripts forward the flags.

8. Self-review checklist



Placeholder scan. No "TBD" / "TODO" / "implement later" in the spec body.



Internal consistency. Architecture diagram, components, data flow, testing all reference the same field names (NetworkProfile, NetworkOverride, CaptureScreenshotOnFailure, ScreenshotPath), the same 8 profile names, the same flag names.



Scope check. Bundled (network sim + screenshot) is appropriate — both are runOne-integration concerns. Not too large for one cycle (1 Containers commit + 1 Lava parent commit).



Ambiguity check. Network condition apply ordering ("after WaitForBoot, before Install/RunInstrumentation") stated in 3 places (component A.3, data flow, testing table). Screenshot capture trigger ("when row.Passed==false AND CaptureScreenshotOnFailure==true") stated in 3 places (component B.1, error handling table, testing table).

Spec status: approved. Plan written inline (next section). Implementation dispatched to a single subagent (smaller scope than Group A-prime / B / pkg-vm so the per-task review pattern collapses to a single implementer + spec/quality review).